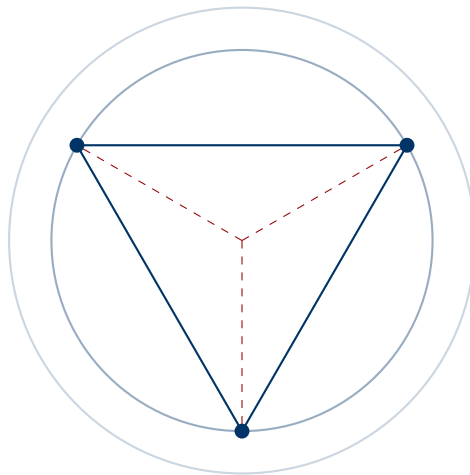


BEYOND THE INCENTER



*A Collection of 50 Selected Geometry Problems
For Modern Olympiads*



Symmetry in Synthetic Proofs

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Exploring the Elegance in Olympiad Geometry.

2026 EDITION

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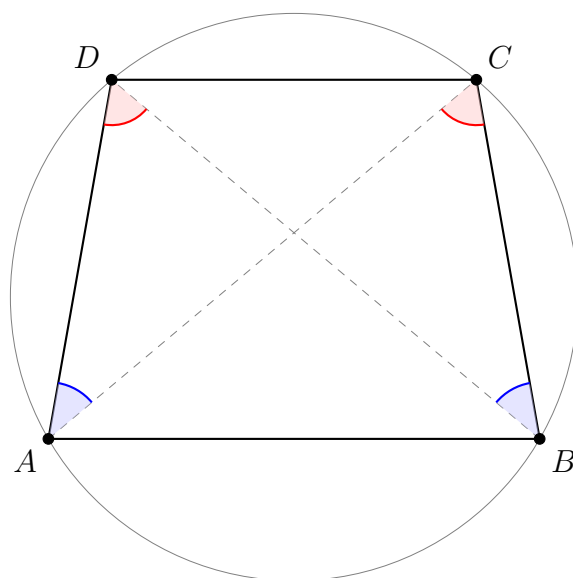
1 Angle Chasing

Invariants in Geometry

Angle chasing is the primary 'language' of Euclidean geometry, and is the fundamental for almost all modern Olympiad Geometry problems. Whether it is the process of identifying equal or supplementary angles, whilst a novice may see angle chasing as basic arithmetic, the experienced would utilise this skill to spot concyclicity, collinearity and concurrency, in order to hunt for 'invariants' in a diagram - properties which are unchanged despite different orientations. Mastery of this skill enables one to simplify a complex diagram into an organised process of slowly turning 'unfamiliar' angles into basic known angles, which we will explore later. However, angle chasing on its own could only get you so far with advanced Geometry. Before looking at some problems, we will first some fundamental theorems or techniques you may have come across.

1.1 Cyclic Quadrilaterals

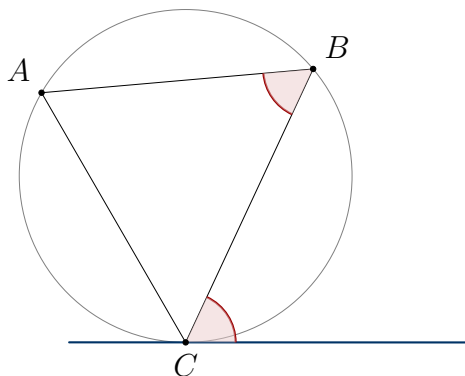
As most of you know from studying it in school curriculum, a set of four points is concyclic if the angles subtended by a side at the other two vertices are equal, or if opposite angles sum to 180° . Just a recap, in the diagram below, $\angle ADB = \angle BCA$, $\angle DAC = \angle CBD$, and $A + C = B + D = 180^\circ$. However, in complex Olympiad problems, these circles are rarely drawn for you; they must be spotted in a cluster of lines in a diagram. This is often the key to proving that two lines are parallel or that a specific point lies on a radical axis. Once a cyclic quadrilateral is identified, it is a signal that you are along the right track to solving the problem. Sometimes, the question gives you a lot of right angles from altitudes or perpendicular bisectors, and this prompts us to search for a pair of right angles in a quadrilateral which instantly suggests it is cyclic. Moreover, cyclic quadrilaterals might invite us to use Ptolemy's Theorem, which will be mentioned in the Geometric Inequalities chapter.



Angles subtended by the same arc are equal.

1.2 Tangency and the Alternate Segment Theorem

Another technique is to be alert when you see tangents. The AST is very often used, and it states that the angle between a chord and the tangent at one of its endpoints is equal to the angle in the alternate segment. By utilising this theorem, one can often transform a problem involving a point of tangency into a standard angle chasing problem, effectively transforming the angles in the triangle onto a straight line.



1.3 Directed Angles

Sometimes, when angle chasing, we realise a proof exists for an 'ordinary' case, for example with acute angled triangles, but fails when a point initially in inside moves to the outside when changing configuration. To combat this, we utilise directed angles, denoted as $\angle(l, m)$. This notation measures the **counterclockwise rotation** required to move line l to line m modulo 180° . By working with directed angles, one ensures that their angle chase remains robust across all possible sketches of the problem.

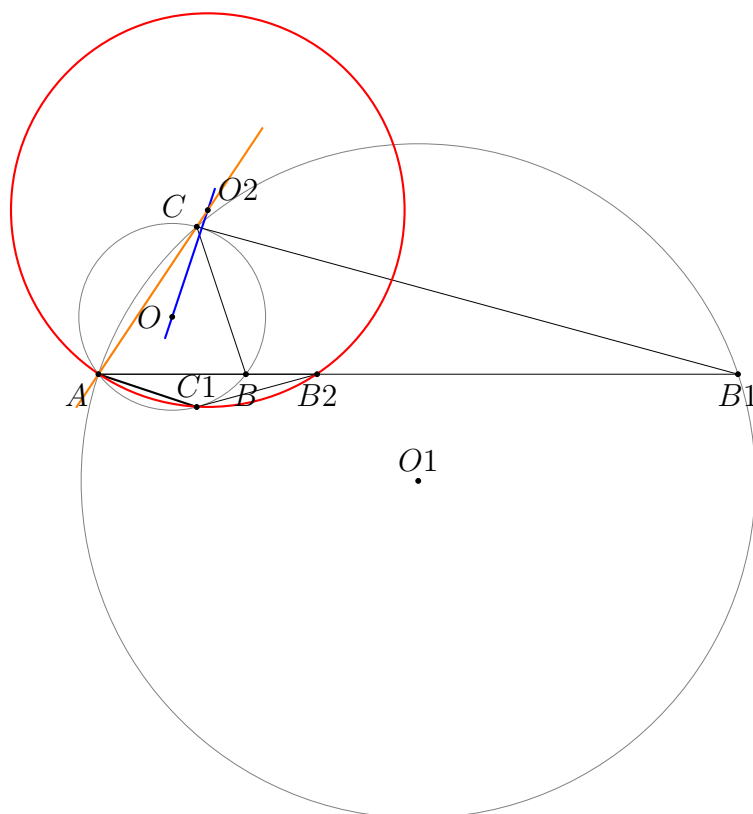
1.4 Reim's Theorem

Oftentimes, we are asked to prove for parallel lines, in which case (although fairly rare) Reim's Theorem provides a sophisticated shortcut. When two circles intersect at two points, any two lines passing through those intersections create a system of angles that can be linked across both circles. If the resulting interior angles are equal, the lines are parallel. This theorem is essentially a double angle chase using cyclic quadrilaterals. Recognizing a Reim configuration can immediately expose the homothety or similarity hidden in the diagram.

Now, here are a few problems with guided hints, solutions and remarks.

Problem 1

Let ABC be a triangle such that its side lengths satisfy $AC > BC > AB$. Let O denote the circumcenter of $\triangle ABC$, and let Γ be its circumcircle. Extend AB past B to B_1 such that the B_1C is tangent to Γ at C . Let O_1 be the circumcenter of $\triangle AB_1C$. AO_1 intersects Γ at a second point C_1 . Let B_2 on AB such that B_2C_1 is tangent to Γ at C_1 . Finally, let O_2 be the circumcenter of $\triangle AB_2C_1$. Given that $OO_2 \perp AO_1$, prove that A, C , and O_2 are collinear.



Hint: Since there are no existing intersections between OO_2 and AO_1 , instead of creating them, we could see the final result as $\angle O_1AC = \angle O_1AO_2$, then work backwards by transforming each of the two angles in to primary angles (i.e. angles in $\triangle ABC$).

Solution: This can be done by rather tedious but pure angle chasing. It suffices to prove $\angle O_1AC = \angle O_1AO_2$.

Note that

$$\begin{aligned} & \angle O_1AC \\ &= 90^\circ - \angle AB_1C \text{ (Removes } O_1) \end{aligned}$$

$$\begin{aligned}
&= 90^\circ - (\angle ABC - \angle BCB_1) \\
&= 90^\circ - \angle ABC - \angle BAC \text{ (Removes } B_1)
\end{aligned}$$

Also,

$$\begin{aligned}
&\angle O_1AO_2 \\
&= 90^\circ - \frac{1}{2}\angle AO_2C_1 \\
&= 90^\circ - \angle B_1B_2C_1 \text{ (Removes } O_2) \\
&= 90^\circ - \angle B_1BC_1 - \angle B_2C_1B \\
&= 90^\circ - \angle B_1BC_1 - \angle BAC_1 \text{ (Removes } B_2) \\
&= 90^\circ - \angle ACE_1 - \angle B_1AO_1 \\
&= 90^\circ - \angle ACB - \angle BCC_1 - \angle B_1AO_1 \\
&= 90^\circ - \angle ACB - 2\angle B_1AS_1 \text{ (Removes } C_1) \\
&= 90^\circ - \angle ACB - 2(90^\circ - \angle ACB_1) \text{ (Removes } O_1) \\
&= 90^\circ - \angle ACB - 180^\circ + 2(180^\circ - \angle ABC) \text{ (Removes } B_1) \\
&= 90^\circ - \angle ABC + \angle BAC \\
&= \angle O_1AC \\
&\implies A_1, O_2, C \text{ are collinear}
\end{aligned}$$

□

Remark

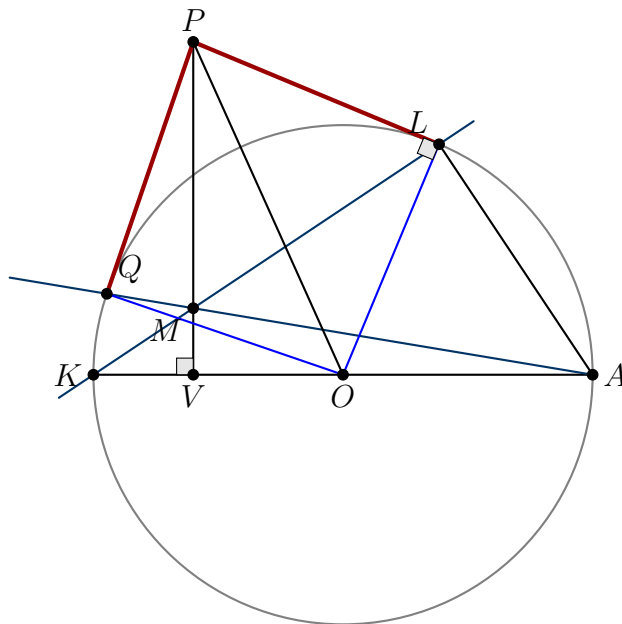
Although this angle chase seemed fairly arduous and time-consuming, we know we will eventually get there as all we used were properties of cyclic quadrilaterals and the alternate segment theorem. We first create order the points in terms of familiarity, and when angle chasing, aim to convert the most unfamiliar points into others via standard processes or theorems we have discussed previously.

In this case:

$$A, B, C \rightarrow O \rightarrow B_1 \rightarrow C_1 \rightarrow B_2 \rightarrow O_2$$

Problem 2

Let ω be a circle with center O , and let AK be a diameter of ω . Let M be a point in the interior of ω not lying on AK . The ray AM intersects ω again at a point Q . Let V be the foot of the perpendicular from M to AK . The tangent to ω at Q intersects the line VM at a point P . Let L be a point on ω ($L \neq Q$) such that PL is tangent to ω . Prove that points K, M , and L are collinear.



Hint: We have PQ and PL are tangents to ω , so alternate segment theorem is key here. Also, if we connect the centre O to Q and L we may have a lot of right angles, which encourages us to search for a pair of opposite right angles to prove concyclicity.

Solution

$$\angle PVO + \angle PLO = 90^\circ + 90^\circ = 180^\circ$$

$\therefore PVOL$ is cyclic

$$\therefore \angle MVL = \angle PVL = \angle POL = \frac{1}{2} \angle QOL = \angle QAL = \angle MAL$$

$\therefore ALMV$ is cyclic

$$\therefore \angle ALM = 90^\circ$$

$$\therefore \angle ALM = \angle ALK = 90^\circ \text{ (as } AK \text{ is the diameter it subtends a right angle)}$$

$\therefore K, L, M$ are collinear

□

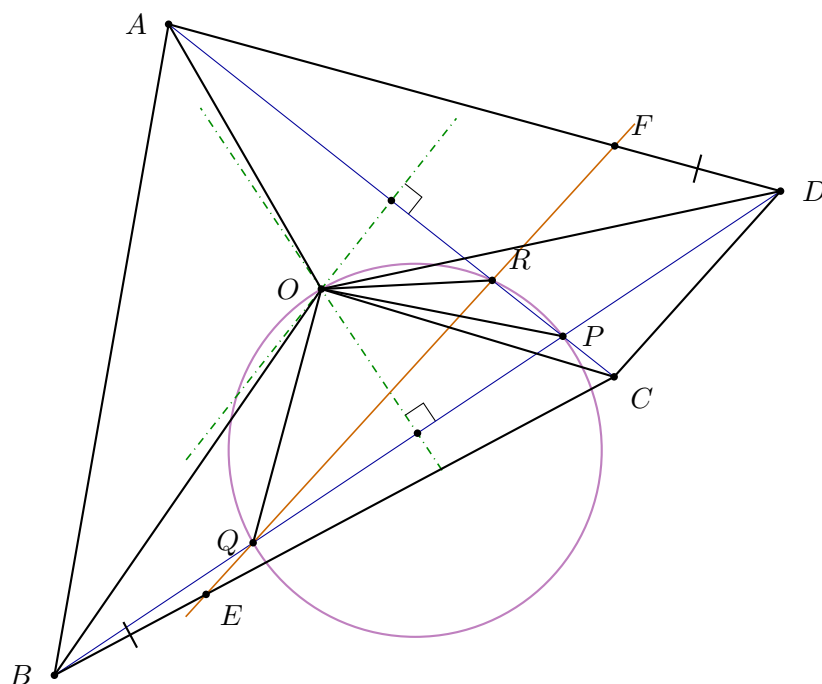
Remark

When proving collinearity, at this stage, angle chasing is rather efficient. It suffices to show $\angle ALM = 90^\circ$, so we cannot use $ALMV$ is cyclic. Instead, we should ignore K and consider M as the intersection of PV and AQ .

Now let's look at a more challenging problem.

Problem 3

Let $ABCD$ be a quadrilateral with $AD = BC$, and let points E and F lie on BC and AD respectively. Let the diagonals AC and BD intersect at P , and let the line EF intersect BD and AC at Q and R respectively. Given that the perpendicular bisectors of AC and BD meet at O , and that O lies on the circumcircle of $\triangle PQR$, prove that $BE = DF$.



Hint: Can you spot any congruent triangles and spiral similarity given the perpendicular bisectors intersect at point O and $AD = BC$?

Solution:

Claim 1:

$$\triangle COA \sim \triangle BOD$$

Since the perpendicular bisectors of AD and BC meet at O , $OB = OD$ and $OC =$

OA . Also, $BC = AD$, so $\triangle OCB \cong \triangle OAD(SSS) \Rightarrow \triangle OCB \sim \triangle OAD$ by spiral similarity.

$$\therefore \angle COA = \angle BOD$$

Since $\triangle COA$ and $\triangle BOD$ are isosceles, they are similar. □

$$\therefore \angle OBD = \angle OCA = \angle OAC = \angle ODB$$

$\Rightarrow BCPO$ and $OADP$ are cyclic.

Claim 2:

$$\triangle OQR \sim \triangle OBC$$

From $OPQR$ cyclic, we know:
$$\begin{cases} \angle OQR = \angle OPR = \angle OBC \\ \angle ORQ = \angle OPQ = \angle OCB \end{cases}$$

$$\Rightarrow \triangle OQR \sim \triangle OBC \sim \triangle ODA$$

□

By rotation:
$$\begin{cases} \triangle OQB \sim \triangle ORC \\ \triangle OQD \sim \triangle ORA \end{cases}$$

Now, by Menelaus:

$$\frac{BE}{CE} = \frac{BQ}{QP} \cdot \frac{PR}{RC}$$

(Removes E)

Similarly,

$$\frac{DF}{AF} = \frac{DQ}{QP} \cdot \frac{PR}{RA}$$

(Removes F)

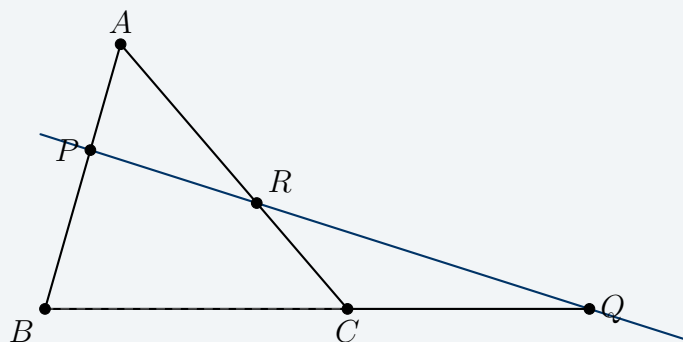
$$\therefore BE = DF \Leftrightarrow \frac{BE}{CE} = \frac{DF}{AF} \Leftrightarrow \frac{BQ}{RC} = \frac{DQ}{RA} \Leftrightarrow \frac{OB}{OC} = \frac{OD}{OA}$$

Since $OB = OD$, $OC = OA$, this holds true. Hence $BE = DF$. □

Remark

Menelaus' Theorem Let $\triangle ABC$ be a triangle. Let a transversal line intersect the lines AB , BC , and CA at points P , Q , and R respectively. The points P, Q, R are collinear **if and only if**:

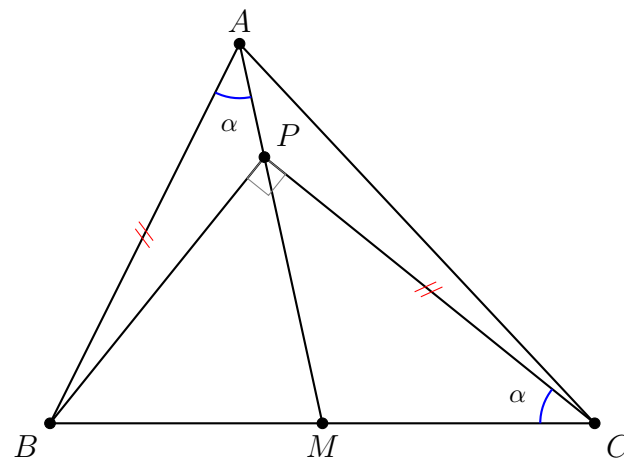
$$\frac{AP}{PB} \cdot \frac{BQ}{QC} \cdot \frac{CR}{RA} = 1$$



Proof by Areas: Notice that

$$\frac{AP}{PB} \cdot \frac{BQ}{QC} \cdot \frac{CR}{RA} = \frac{[AQR]}{[BQR]} \cdot \frac{[BQR]}{[CQR]} \cdot \frac{[CQR]}{[AQR]} = 1$$

and we are done. □



2 The Five Centers of a Triangle

3 Power of a Point and Radical Axis

4 Harmonic Lines and Points

5 Complete Quadrilaterals

6 Geometric Transformations

7 Trigonometric Methods

8 Complex Numbers in Geometry

9 Geometric Inequalities

Problem 1

Let $ABCD$ be a trapezoid with $AB \parallel CD$. Let ω_1 be a circle tangent to the sides DA, AB , and BC , with center O_1 . Let ω_2 be a circle tangent to the sides BC, CD , and DA , with center O_2 . Let P be the point of tangency of ω_1 with AB , and Q be the point of tangency of ω_2 with CD . Prove that the lines AC, BD , and PQ are concurrent.

Solution: Let AC, BD intersect PQ at E_1 and E_2 respectively. So it suffices to prove E_1 and E_2 are the same point.

$$\begin{aligned}
 &\Leftrightarrow \frac{PE_1}{E_1Q} = \frac{PE_2}{E_2Q} \\
 &\Leftrightarrow \frac{AP}{QC} = \frac{BP}{DQ} \text{ (Removes } E_1, E_2) \\
 &\Leftrightarrow \frac{AP}{BP} = \frac{CQ}{DQ} \\
 &\Leftrightarrow \frac{r_1 \cdot \frac{1}{2} \cot \angle BAD}{r_1 \cdot \frac{1}{2} \cot \angle ABC} = \frac{r_2 \cdot \frac{1}{2} \cot \angle BCD}{r_2 \cdot \frac{1}{2} \cot \angle ADC} \text{ (Removes } P, Q) \\
 &\Leftrightarrow \frac{\cot \frac{\angle BAD}{2}}{\cot \frac{\angle ABC}{2}} = \frac{\tan \frac{\angle ABC}{2}}{\tan \frac{\angle BAD}{2}} \\
 &\Leftrightarrow \frac{\cot \frac{\angle BAD}{2}}{\cot \frac{\angle ABC}{2}} = \frac{\tan \frac{\angle ABC}{2}}{\tan \frac{\angle BAD}{2}}
 \end{aligned}$$

This is trivial as $\tan$ and $\cot$ are reciprocals of each other. □

Problem 3

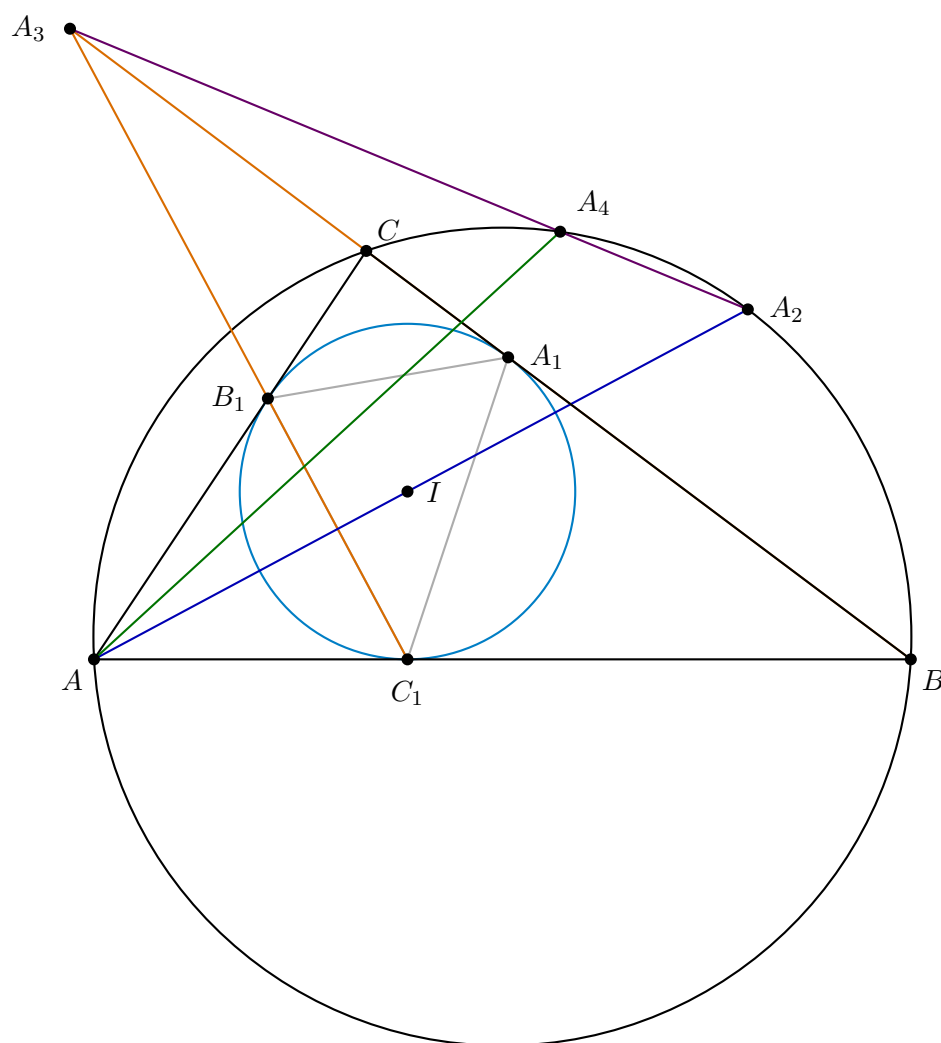
Let ABC be a triangle with incenter I . Let the incircle of $\triangle ABC$ be tangent to the sides BC, CA, AB at points A_1, B_1, C_1 , respectively.

Let the extension of AI meet the circumcircle of $\triangle ABC$ at point A_2 .

Let A_3 be the intersection of the lines BC and B_1C_1 .

Let the line A_2A_3 intersect the circumcircle of $\triangle ABC$ at a second point A_4 .

Points B_4 and C_4 are constructed in a similar way starting from vertices B and C , respectively. Prove that the lines AA_4, BB_4 , and CC_4 are concurrent.



Solution:

AA_4, BB_4, CC_4 are concurrent

$\Leftrightarrow \prod \frac{BA_4}{A_4C} = 1$ (Ceva's Theorem)

$\Leftrightarrow$ Remove A_4 :

Since $\angle A_3BA_4 = \angle A_3A_2C$ as BCA_2A_4 is cyclic, we get

$$\triangle A_3A_4B \sim \triangle A_3CA_2(AA)$$

$$\Rightarrow \frac{BA_4}{A_2C} = \frac{A_3B}{A_3A_2}$$

It follows that:

$$\triangle A_3A_4C \sim \triangle A_3BA_2(AA)$$

$$\Rightarrow \frac{CA_4}{A_2B} = \frac{A_3C}{A_3A_2}$$

Dividing the two equations, we obtain:

$$\frac{BA_4}{A_4C} = \frac{A_3B}{A_3C} \cdot \frac{A_2C}{A_2B}$$

Now, according to the Incenter-Excenter Lemma, we have $A_2C = A_2B = A_2I$, so it suffices to prove $\prod \frac{A_3B}{A_3C} = 1$, which can be done using Menelaus. We now use Menelaus' Theorem for $\triangle ABC$ and line $C_1B_1A_3$:

$$\prod \left(\frac{BC_1}{C_1A} \cdot \frac{AB_1}{B_1C} \right) = 1$$

Note $AC_1 = AB_1$, hence:

$$\prod \frac{BC_1}{B_1C} = 1$$

This is trivial as:

$$\frac{\prod BC_1}{\prod B_1C} = \frac{BC_1 \cdot CA_1 \cdot AB_1}{B_1C \cdot A_1B \cdot C_1A} = 1$$

□

Video Explanation of Incenter-Excenter Lemma - Kevin Hu

Let ABC be an acute triangle with circumcircle Γ and circumcenter O . Let M be the midpoint of the side BC . Let H be the orthocenter of $\triangle ABC$, and let D, E , and F be the feet of the perpendiculars from vertices A, B , and C to the opposite sides BC, CA , and AB , respectively.

Solution:

A, Z, M are collinear

$$\Leftrightarrow \frac{[\triangle XAM]}{[\triangle YAM]} = 1 \text{ (removes } Z)$$

$$\Leftrightarrow \frac{AX \cdot AM \cdot \sin \angle XAM}{AY \cdot AM \cdot \sin \angle YAM} = 1 \text{ (Sine Rule)}$$

$$\Rightarrow \frac{AX \cdot \sin \angle XAM}{AY \cdot \sin \angle YAM} = 1$$

We have $\begin{cases} \angle EAX = \angle BAY = \angle BAO = \frac{\pi}{2} - C \\ \angle AEX = \angle ABY = \frac{\pi}{2} \end{cases}$ (Orthic $\triangle DEF$ suggests $\triangle AEF \sim \triangle ABC$ as $BFEC$ is cyclic.)

$$\Rightarrow \triangle AEX \sim \triangle ABY \text{ (AA)}$$

$$\Rightarrow \frac{AX}{AY} = \frac{AE}{AB}$$

$$\Rightarrow \frac{AE}{AB} \cdot \frac{\sin \angle DAM}{\sin \angle OAM} = 1 \text{ (Removes } X, Y)$$

Note that:

$$\frac{\sin \angle OAM}{\sin \angle DAM} = \frac{\sin \angle OAG}{\sin \angle GAH} = \frac{OG}{GH} \cdot \frac{AH}{AO} = \frac{1}{2} \cdot \frac{2R \cdot \cos A}{R} = \cos A$$

(Note that G represents the centroid of $\triangle ABC$)

$$\Rightarrow \frac{AE}{EB} = \cos A \text{ which is trivial.}$$

Euler's Line

In any non-equilateral triangle, the circumcenter O , the centroid G , and the orthocenter H are collinear. The line passing through these points is called the *Euler Line*. Furthermore, G lies between O and H such that:

$$HG = 2GO$$

This relationship is elegantly demonstrated by reflecting H across the midpoint M of side BC to a point H' , which forms a parallelogram $BHCH'$ and identifying AH' as a diameter of the circumcircle passing through O . Since OM is a midline of $\triangle AHH'$, it follows that $AH = 2OM$ and $AH \parallel OM$; combined with the fact that the centroid G divides the median AM in a $2 : 1$ ratio, the triangles $\triangle AGH$ and $\triangle MGO$ are similar by *SAS*.

Video Explanation of Euler's Line - Kevin Hu

□

Alternative:

One may find the following auxiliary line:

Double the length of HM to H' , so by congruent triangles and midpoints, $CH'BH$ is a parallelogram.

$$\therefore \angle BH'C = \angle BHC = \pi - A$$

$$\therefore ABH'C \text{ is cyclic}$$

Now $H'B \parallel HC$ and $HC \parallel AB$

$$\therefore H'B \perp AB$$

$$\therefore A, D, H' \text{ are collinear and } AH' = 2R$$

We know Z is the midpoint of XY and M is the midpoint of HH' .

So it suffices to show $XY \parallel HH'$, or in other words

$$\Leftrightarrow \frac{AX}{XH} = \frac{AY}{YH'}$$

From $\triangle AHE \sim \triangle AH'B$, we know X and Y are corresponding points of similarity.

Problem 5

For positive integer $n \geq 2$, let there be positive real numbers $x_1 < x_2 < \dots < x_n$, $y_1 < y_2 < \dots < y_n$. Prove that for any $C \in (-2, 2)$,

$$\sum_{i=1}^n \sqrt{x_i^2 + Cx_i y_i + y_i^2} < \sum_{i=1}^n \sqrt{x_i^2 + Cx_i y_{i+1} + y_{i+1}^2},$$

of which $y_{n+1} = y_1$.